

HYDROFARMS AGRI SOLUTIONS

MAIN ELECTRICAL POWER DISTRIBUTION,
EARTHING AND EMERGENCY GENERATOR

Activity-Specific Installation, Testing and Commissioning Method Statement

Document field	Controlled value
Document code	HF-GH01-MST-ELC-001
Revision	00 - Draft for coordination
Project	One-Feddan NFT Lettuce Greenhouse Project
Location	Alexandria, Egypt
Main supply basis	250 kVA; 400 A incomer; 400/230 V, 3 phase, 50 Hz
Emergency source	150 kVA diesel generator for critical loads only
Prepared by	Hydrofarms Agri Solutions
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NON-NEGOTIABLE ROUTING RULE
All cables are exposed on covered galvanized trays or protected exposed conduit. No electrical cable is buried anywhere in the project scope.

1. Purpose, Scope and Boundaries

This method controls installation, termination, earthing, testing, energization and emergency operation of the greenhouse LV system, from utility entry and generator through all project distribution boards and external equipment feeders.

Design item	Adopted requirement
Supply	400/230 V, 3 phase, 4 wire, 50 Hz; TN-S downstream of the project main earth bar.
Project design basis	250 kVA preliminary utility capacity; proposed 400 A main incomer.
Incoming feeder	30 m provisional visible route: 4C x 300 mm ² Al XLPE/SWA/PVC 0.6/1 kV plus 1C x 150 mm ² Cu ECC.
Emergency source	150 kVA, 400 V diesel generator with ATS/AMF and a separate emergency distribution board.
Cable installation	All cables exposed on covered galvanized trays or protected exposed conduit. No buried electrical cables.
Reference cable supplier	EI Sewedy Cables or approved equal, with project submittal and test certificates.
Environmental duty	Wet, humid, fertilizer-bearing greenhouse service; UV, corrosion and wash-down exposure considered.
Enclosures	MDB IP55 minimum in protected location; field DBs, boxes and isolators IP65 minimum.
Earthing	Three earth groups, each with three 16 mm x 3 m copper-bonded rods; total nine rods; target resistance ≤ 5 ohm.
Surge/lightning	Type 2 SPD at MDB. No independent lightning-protection system in this package; SPD is not a substitute for one.

Package boundary: internal control panels of the five chillers, WTP and Geneina fertigation skid remain supplier scope. This package provides external power, protection, isolation, earthing and interface terminals.

2. Applicable Standards and Submittals

- IEC 60364 series and IEC 60364-6 for verification.
- IEC 61439-1/-2 for LV assemblies.
- IEC 60502-1 for 0.6/1 kV XLPE power cables.
- IEC 60947 series and IEC 60529 for devices and IP ratings.
- ISO 8528 series for the 150 kVA generator.
- Egyptian code, utility requirements and approved manufacturer instructions; the stricter requirement governs.

Submittals: SLD, load/fault/protection and voltage-drop studies, panel GA, cable/tray and earthing layouts, generator/ATS logic, interface matrix and FAT procedures.

3. Design Load Schedule

Code	Load	Qty x kW	Connected	Peak	Basis
LD-01	Production exhaust fans	35 x 1.50	52.50	52.50	All may run at summer peak
LD-02	Service exhaust fans	3 x 1.50	4.50	4.50	Staged 1 + 2
LD-03	Production circulation fans	28 x 0.37	10.36	10.36	800 mm fans
LD-04	Service circulation fans	2 x 0.25	0.50	0.50	600 mm fans
LD-05	Production NFT pumps	3 x 5.50	16.50	11.00	2 duty + 1 shared standby
LD-06	Nursery NFT pumps	2 x 1.50	3.00	1.50	1 duty + 1 standby
LD-07	Reserve manoeuvre pumps	2 x 2.20	4.40	2.20	1 duty + 1 standby
LD-08	Germination pump	1 x 0.37	0.37	0.37	Service greenhouse
LD-09	Production pad pumps	5 x 2.20	11.00	8.80	4 duty + 1 spare
LD-10	Service pad pumps	2 x 0.75	1.50	0.75	1 duty + 1 spare
LD-11	Production nutrient chillers	4 x 10.00	40.00	40.00	Four 7.5 TR units; provisional input
LD-12	Nursery nutrient chiller	1 x 10.00	10.00	10.00	One 7.5 TR unit; provisional input
LD-13	Water-treatment station	1 x 11.00	11.00	0.00	Scheduled outside peak where practical
LD-14	Fertigation package	1 x 2.00	2.00	2.00	External feeder only
LD-15	Germination lighting	1 x 1.50	1.50	1.50	Production relies on sunlight
LD-16	Packing/service small power	1 x 5.00	5.00	3.00	Sockets and packing equipment
LD-17	Controls and sensors	1 x 1.00	1.00	1.00	Simple local controls
LD-18	Internal/external shade motors	1 x 5.00	5.00	0.00	Intermittent operation
LD-19	External/service lighting	1 x 3.00	3.00	1.50	Allowance
TOTAL	Calculated design total	-	183.13	151.48	Before power factor and diversity confirmation

The approved cooling basis is five chillers at 7.5 TR each: four production and one nursery. A provisional 10 kW input per unit is used; supplier MCA/MOCP and starting data govern final protection and termination.

4. Distribution Architecture and Panels

Panel	Qty	Duty	Minimum requirement
MDB	1	400 A Form 2B or better	Metering, outgoing MCCBs, neutral/earth bars, Type 2 SPD and 20% spare space.
EMG-DB	1	150 kVA emergency system	Selected fan stages, duty NFT/pad pumps, controls, alarms and essential services.
FAN-DB-PROD	1	35 exhaust fans	Four operating sectors; outgoing group circuits to 14 bay boxes.
FAN-DB-SERV	1	3 exhaust fans	Independent 1 + 2 staging.
CIRC-DB	1	30 circulation fans	Grouped Manual-Off-Auto control.
PAD-DB	1	Pad pumps	Production and service pump protection with level and climate interlocks.
NFT-DB	1	NFT/manoeuvre pumps	Duty/standby logic and chiller permissive interfaces.
CH-DB	1	5 x 7.5 TR chillers	Five protected outgoing feeders; vendor internal panels excluded.
WTP-DB	1	WTP interface	External power and isolation only; package panel excluded.
FER-DB	1	Fertigation interface	External power and isolation only; Geneina internal panel excluded.
SV-DB	1	Service greenhouse	Germination, packing small power and service lighting feeders.
SHADE-DB	1	Screen motors	Power/interface only; supplier motors, gearboxes and mechanism excluded.
CTRL-DB	1	Control supply	Protected control supply and interface terminals; automation devices in separate chapter.

5. Main Feeder Cable Schedule

Code	Route	Power cable	Separate ECC	m
CB-001	Utility/meter -> MDB	4C x 300 mm2 Al XLPE/SWA/PVC	1C x 150 mm2 Cu	30
CB-002	MDB -> FAN-DB-PROD	4C x 70 mm2 Cu XLPE/SWA/PVC	1C x 35 mm2 Cu	25
CB-003	MDB -> FAN-DB-SERV	4C x 10 mm2 Cu XLPE/SWA/PVC	1C x 10 mm2 Cu	40
CB-004	MDB -> CIRC-DB	4C x 16 mm2 Cu XLPE/SWA/PVC	1C x 16 mm2 Cu	60
CB-005	MDB -> PAD-DB	4C x 16 mm2 Cu XLPE/SWA/PVC	1C x 16 mm2 Cu	50
CB-006	MDB -> NFT-DB	4C x 50 mm2 Cu XLPE/SWA/PVC	1C x 25 mm2 Cu	35
CB-007	MDB -> CH-DB	4C x 50 mm2 Cu XLPE/SWA/PVC	1C x 25 mm2 Cu	40
CB-008	MDB -> WTP-DB	4C x 16 mm2 Cu XLPE/SWA/PVC	1C x 16 mm2 Cu	45
CB-009	MDB -> FER-DB	4C x 6 mm2 Cu XLPE/SWA/PVC	1C x 6 mm2 Cu	30
CB-010	MDB -> SV-DB	4C x 25 mm2 Cu XLPE/SWA/PVC	1C x 16 mm2 Cu	105
CB-011	MDB -> CTRL-DB	5C x 6 mm2 Cu XLPE/PVC	1C x 6 mm2 Cu if required	25
CB-012	MDB -> SHADE-DB	4C x 16 mm2 Cu XLPE/SWA/PVC	1C x 16 mm2 Cu	60
CB-013	Generator -> ATS/AMF	4C x 95 mm2 Cu XLPE/SWA/PVC	1C x 50 mm2 Cu	15
CB-014	ATS/AMF -> EMG-DB	4C x 95 mm2 Cu XLPE/SWA/PVC	1C x 50 mm2 Cu	15

6. Field Cabling and Containment

Code	Circuit group	Cable	Qty	Basis
CB-015	Production fan bay feeders	4C x 6 mm2 Cu + 1C x 6 mm2 ECC	485 m	14 IP65 bay boxes
CB-016/17	Exhaust fan final drops	4C x 2.5 mm2 Cu + ECC	355 m	35 production + 3 service
CB-018	Circulation fan feeds	4C x 2.5 mm2 Cu + ECC	365 m	30 fans
CB-019/20	Cooling-pad pump feeds	4C x 4 / 2.5 mm2 Cu + matching ECC	210 m	Production and service
CB-021/23	NFT/manoeuvre pump feeds	4C x 6 / 4 / 2.5 mm2 Cu + ECC	170 m	Seven pump connections
CB-024/25	Five 7.5 TR chiller feeds	4C x 10 mm2 Cu + 1C x 10 mm2 ECC	115 m	Final MCA/MOCP check mandatory
CB-026/27	Fertigation/germination pump	4C or 3C x 2.5 mm2 Cu + ECC	50 m	External feeder only
CB-028	Germination lighting feeder/circuits	3C x 2.5 mm2 Cu	120 m	Final circuits in lighting chapter
CB-029	Shade motor power	4C x 2.5 mm2 Cu in tray/conduit	780 m	Internal and external zones
CB-030	Control cables	4C/7C x 1.5 mm2 Cu	850 m	Commands, floats and feedback
Containment		Quantity	Use	Installation requirement
300 mm HDG ladder tray		230 m	Main power routes	Covered, heavy duty, bonded at every joint.
200 mm HDG perforated tray		390 m	Branch power routes	Covered, supported and corrosion protected.
100 mm HDG tray/protected conduit		260 m	Control and small power	Segregated from power where required.

7. Earthing, Bonding and Surge Protection

Install three earth groups, each comprising three copper-bonded rods minimum 16 mm diameter x 3 m long, with one accessible inspection point per group. Bond all panels, generator, trays, greenhouse steel, equipment frames, pumps, chillers and metallic services.

- A separate copper ECC accompanies each main feeder; SWA is not the only protective conductor.
- Bond tray joints, flexible connections and structural discontinuities.
- Combined target earth resistance is ≤ 5 ohm; add rods only if testing requires.
- Provide a 3P+N Type 2 SPD at MDB. No independent lightning-protection system is included.

8. Emergency Generator and Load-Shedding Philosophy

Load group	Emergency selection	Running	Control rule
Production exhaust	Up to 18 fans in staged groups	27.0 kW	Start group-by-group
Service exhaust	Up to 2 fans	3.0 kW	Start after production minimum
Circulation	Selected 10 production fans	3.7 kW	Optional by temperature
NFT circulation	2 production + 1 nursery duty pumps	12.5 kW	Highest process priority
Pad cooling	4 production + 1 service duty pumps	9.55 kW	Before corresponding fans
Controls/fertigation	Controllers, alarms and fertigation feeder	3.0 kW	Continuous essential
Emergency lighting/small power	Selected circuits	3.0 kW	Non-process essential
TOTAL	Normal emergency operating selection	61.75 kW	Motor-starting sequence governs

Chillers, WTP, packing loads and shade drives are normally shed. Start controls, NFT pumps, pad pumps, fan groups and selected circulation fans in sequence with 5-10 seconds between major motor groups. ATS interlocking shall prevent utility-generator paralleling and backfeed.

9. Installation Sequence

- Approve the single-line diagram, load schedule, discrimination study, voltage-drop schedule, tray routing and equipment interface schedule before procurement.
- Release plinths, wall zones and structural support points. Confirm that electrical routes remain above ground and do not obstruct fans, pads, screens, doors or maintenance access.
- Install the generator base, MDB/ATS/EMG-DB and sub-DBs plumb and level; maintain front, rear and ventilation clearances required by the manufacturer.
- Install HDG tray supports first, then trays, covers, bends, tees, expansion provisions and earth-bonding jumpers. Remove sharp edges and restore damaged galvanizing.
- Pull power cables before control cables. Observe pulling tension and bend radius; segregate circuits, support vertical drops and seal all field entries with IP-rated glands.
- Terminate conductors with calibrated crimping tools, bimetallic lugs at aluminium-to-copper interfaces, ferrules for control conductors and heat-shrink phase identification.
- Install 14 production fan bay boxes and one service fan box. Verify each bay circuit, local lockable isolator and fan rotation before enabling automatic staging.
- Complete three earth groups, bond all trays, panels, generator, equipment frames and greenhouse steel, then test resistance before energization.
- Complete dead tests, protection settings and point-to-point checks. Energize progressively from MDB to sub-DBs and then individual loads under an approved switching programme.

10. Testing, Commissioning and Acceptance

Test	Method	Acceptance
Material/identity	Certificates, ratings, IP class, cable drums and panel data	Approved submittal and traceability
Panel routine verification	IEC 61439 routine checks, wiring, torque and functional inspection	Signed panel-builder record
Insulation resistance	Power and control circuits isolated from electronics	Acceptable to governing code/manufacture
Protective continuity	ECC, tray bonds, frames and equipment	Continuous low-resistance protective path
Earth resistance	Three-point/fall-of-potential test per earth group	Combined project value ≤ 5 ohm
Polarity/phase sequence	All final circuits and three-phase boards	Correct polarity and clockwise sequence
Protection settings	MCCB/MPCB/overload/short-circuit coordination	Matches approved study and motor nameplates
Motor rotation	Bump test each pump, fan and drive uncoupled where required	Correct airflow/flow direction
ATS/AMF	Utility failure, generator start, transfer, retransfer and cooldown	Automatic sequence without backfeed
Emergency load step	Apply approved load sequence and record V, Hz, current and smoke	Stable operation without overload or excessive dip
Integrated interlocks	Low level, trip, climate command, E-stop and remote status	Cause-and-effect fully demonstrated

10.1 Inspection and Test Plan

No.	Inspection stage	Method	Type	Record
01	SLD, load/protection study and routes	Document review	H	Approved design pack
02	Panels and generator before dispatch	FAT/witness	W	FAT certificates
03	Supports, trays and bonding before cabling	Visual/measurement	H	Route release
04	Cables, glands and terminations	Identity/torque	W	Cable schedule
05	Earthing before energization	Continuity/resistance	H	Earth report
06	Dead electrical tests	Instrumented	H	Test sheets
07	Progressive energization and rotation	Functional	H	Energization record
08	ATS/generator emergency-load test	Performance	H	Generator report
09	Integrated controls and alarms	Cause-and-effect	W	SAT record
10	As-builts, labels, training and spares	Record review	R	Turnover dossier

Intervention codes: H = Hold, W = Witness, S = Surveillance, R = Record review.

10.2 Hold points

- HP-ELC-01: Design and protection studies accepted before procurement.
- HP-ELC-02: Containment and bonding accepted before cable pulling.
- HP-ELC-03: Terminations and earthing accepted before energization.
- HP-ELC-04: Dead tests and settings accepted before first energization.
- HP-ELC-05: ATS, generator load-step and integrated tests accepted before handover.

11. HSE, Records and Handover

Hazard	Mandatory control	Residual risk
Live electricity	Permit, LOTO, test-before-touch and authorized electricians	Medium
Cable pulling	Rated rollers/grips, communication and controlled pull zone	Low
Work at height	Approved access, fall protection and exclusion zone	Medium
Generator	Ventilation, exhaust guarding, fire and spill controls	Medium
Wet environment	IP equipment, RCD where required, bonding and dry testing	Low
Unexpected start	Local isolation, control inhibit and guarding	Low

Handover includes SLDs, panel/cable schedules, settings, FAT/SAT, IR/continuity/earth tests, torque records, generator/ATS report, certificates, as-builts, labels, O&M manuals, warranties, spares and training.